Multiplying and Dividing Integers

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Integers

Integers are positive whole numbers, negative whole numbers, and zero.

$$\ldots, -8, -3, 0, 4, 12, \ldots$$

Two Steps

When multiplying or dividing integers:

Step 1: Decide the **sign of the answer**.

Step 2: Multiply or divide the **absolute values**.

Sign Rules: Multiplication

Positive $\times$ Positive $=$ **Positive**

Negative $\times$ Negative $=$ **Positive**

Positive $\times$ Negative $=$ **Negative**

Negative $\times$ Positive $=$ **Negative**

Sign Rules: Division

The same sign rules apply to division.

$$\text{Positive} \div \text{Positive} = \textbf{Positive}$$

$$\text{Negative} \div \text{Negative} = \textbf{Positive}$$

$$\text{Positive} \div \text{Negative} = \textbf{Negative}$$

$$\text{Negative} \div \text{Positive} = \textbf{Negative}$$

Key Idea

Same signs give a **positive** answer.

Different signs give a **negative** answer.

Memory Trick

Think of **friends** and **enemies**.

- Friend of a friend is a friend.
  $(+)(+) = +$
- Enemy of an enemy is a friend.
  $(-)(-) = +$
- Friend of an enemy is an enemy.
  $(+)(-) = -$
- Enemy of a friend is an enemy.
  $(-)(+) = -$

Zero Rule for Multiplication

The product of any integer and zero is zero.

$$(-9)(0) = 0$$

$$0(15) = 0$$

Zero Rule for Division

Zero divided by any nonzero integer is zero.

$$0 \div (-4) = 0$$

However, division by zero is **undefined**.

$7 \div 0$ is undefined

Examples

Example 1

Find $(-5)(3)$.

Different signs, so the answer is negative.

$$5 \times 3 = 15$$

$$(-5)(3) = -15$$

Example 2

Find $(-4)(-6)$.

Same signs, so the answer is positive.

$$4 \times 6 = 24$$

$$(-4)(-6) = 24$$

Example 3

Find $36 \div (-9)$.

Different signs, so the answer is negative.

$$36 \div 9 = 4$$

$$36 \div (-9) = -4$$

Example 4

Find $(-48) \div (-6)$.

Same signs, so the answer is positive.

$$48 \div 6 = 8$$

$$(-48) \div (-6) = 8$$

Practice Exercises

Practice 1

Find the product or quotient of the pair of integers.

$$(-6)(4)$$

Practice 2

Find the product or quotient of the pair of integers.

$$(-8)(-7)$$

Practice 3

Find the product or quotient of the pair of integers.

$$9(-5)$$

Practice 4

Find the product or quotient of the pair of integers.

$$(-12)(3)$$

Practice 5

Find the product or quotient of the pair of integers.

$$(-11)(-6)$$

Practice 6

Find the product or quotient of the pair of integers.

$$42 \div (-7)$$

Practice 7

Find the product or quotient of the pair of integers.

$$(-64) \div (-8)$$

Practice 8

Find the product or quotient of the pair of integers.

$$(-81) \div 9$$

Practice 9

Find the product or quotient of the pair of integers.

$$100 \div (-10)$$

Practice 10

Find the product or quotient of the pair of integers.

$$(-72) \div (-9)$$

Activities

Activity 1

Solve.

$$(-7)(5)$$

Activity 2

Solve.

$$(-9)(-4)$$

Activity 3

Solve.

$$8(-6)$$

Activity 4

Solve.

$$(-14)(2)$$

Activity 5

Solve.

$$(-12)(-5)$$

Activity 6

Solve.

$$56 \div (-8)$$

Activity 7

Solve.

$$(-72) \div (-9)$$

Activity 8

Solve.

$$(-96) \div 12$$

Activity 9

Solve.

$$120 \div (-12)$$

Activity 10

Solve.

$$(-90) \div (-10)$$

Thank you!